

Sabarinath Sri Dhandapani

Aerospace Engineer

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PROFILE

Aerospace engineer with an MSc from the University of Leicester, working across CFD, flight dynamics, control system design, and thermal analysis. Built a 7-state F-16 longitudinal flight dynamics model in MATLAB/Simulink and designed an LQR altitude-hold controller around it, verifying performance held up under a +20% perturbation in the aircraft dynamics before trusting the result. Separately, used ANSYS Fluent to estimate propulsion performance across multiple configurations and correlated simulation predictions against physical wind-tunnel test data. Writes things down properly — technical reports, documented assumptions, results that hold up when someone else checks them. Looking for an entry-level engineering role in CFD, flight dynamics, control systems, or thermal analysis within civil aerospace, motorsport, or related high-performance engineering sectors.

EDUCATION

MSc Aerospace Engineering

2024 – 2025

University of Leicester, UK

- Specialised in aerodynamic modelling, flight dynamics, computational simulation, and control systems engineering.
- Core modules: Advanced Fluid Dynamics (turbulence modelling theory, boundary layer theory, heat transfer and thermal analysis), Computational Methods, Systems Modelling, Flight Mechanics, Control Engineering.

BEng (Hons) Aeronautical Engineering

2019 – 2023

Kumaraguru College of Technology, India — GPA 3.2/4.0 (UK equivalent: Upper Second-Class Honours, 2:1)

- Core strengths: aerodynamics, structures, propulsion systems, CFD.
- Leadership: Treasurer — KI Aero Modelling Club; Event Lead — SEDS KCT.

TECHNICAL SKILLS

- Simulation & Control: MATLAB/Simulink, LQR control design, state-space modelling, dynamic and steady-state system analysis
- CFD & Thermal Analysis: ANSYS Fluent (MRF), STAR-CCM+/Simcenter, COMSOL — thrust estimation, aerodynamic loading, heat transfer, conjugate heat transfer, fluid-structure interaction, aeroelasticity
- CAD & Design: CATIA, SOLIDWORKS (CSWP Certified) — mechanical design, complex assembly modelling, parametric part design, structural and aerospace design modelling
- Systems Engineering: Requirements definition, verification & validation (V&V), performance analysis, data correlation
- Programming: Python, MATLAB, C — simulation, data processing, automation
- Embedded & Hardware: Arduino, Proteus, STK, OpenRocket
- Documentation: LaTeX, MS Office — technical reporting and presentation

PROJECT EXPERIENCE

Altitude-Hold Autopilot Design for a Fixed-Wing Fighter Aircraft Model

May 2025 – Sep 2025

University of Leicester — MSc Research Project

- Built a 7-state linear state-space model of F-16 longitudinal dynamics (altitude, pitch angle, forward velocity, angle of attack, pitch rate, plus two further stability states) in MATLAB/Simulink, with throttle and elevator as the two control inputs; augmented it with an integral altitude-error state (8 states total) and confirmed full controllability (rank 8) before designing the controller.
- Designed a full-state-feedback LQR controller with integral action, tuning the Q/R weighting matrices to trade off tracking speed against control effort.
- Achieved a 100 ft step response with 0.9% overshoot, 5.74 s settling time (2% band), and 0.023 ft steady-state error in the nominal case.
- Stress-tested the design against a +20% increase in the state matrix A: settling time extended to 7.22 s and overshoot fell to 0%, with steady-state error still just 0.019 ft — confirming the controller held stability and tracking accuracy outside the nominal design point.

LiDAR-RADAR Sensor Fusion for Moving Target Detection

Aug 2022 – Apr 2023

Kumaraguru College of Technology — Undergraduate Team Project (3-person team)

- Owned the Kalman filter design and MATLAB simulation workstream for the team, building a nonlinear coordinated-turn motion model to fuse LiDAR and RADAR measurements into a single state estimate.
- Implemented and directly compared an EKF and UKF under identical simulated trajectories rather than assuming which would perform better.
- The fused UKF achieved 0.41 m position RMSE and 0.36 m/s velocity RMSE — 34.9% better than LiDAR-only and 47.4% better than RADAR-only tracking; the fused EKF trailed at 0.49 m / 0.49 m/s.
- The UKF's higher per-update cost (0.58 ms vs. 0.31 ms) stayed well inside the 50 ms sampling budget, so the accuracy gain came at negligible real-time cost.

Dynamic Thrust Estimation of a Fixed-Wing UAV

Jan 2022 – Jun 2022

Kumaraguru College of Technology — Undergraduate Project

- Built a 3D Moving Reference Frame (MRF) CFD model in ANSYS Fluent to estimate propeller thrust, resolving rotating-zone boundaries and blade geometry rather than using a simplified actuator-disk approximation.
- Ran a mesh-independence study (1.18M/2.36M/4.72M cells), confirming convergence to within 1.8% before trusting the result — the same verify-before-trust habit as the LQR robustness testing.
- At the design point (6,800 rpm, 16 m/s), the model predicted 18.69 N thrust and 0.79 N·m torque; compared three propulsion configurations and selected the 3-blade/305 mm design for a thrust-to-drag ratio of 1.42 versus 1.32 and 1.28 for the alternatives.

Metal-Detection Security Drone — Structural Design

Aug 2021 – Dec 2021

Kumaraguru College of Technology — Undergraduate Project

- Designed a three-arm tricopter airframe in CATIA around an integrated metal-detection payload, iterating arm geometry, central-body layout, and reinforcement across four design revisions.
- Sized the structure against five load cases (hover, 2.5g manoeuvre, asymmetric thrust, landing, payload offset); the landing case governed at 72.1 MPa peak stress with a minimum factor of safety of 2.07.
- Rejected a lighter fourth iteration (0.44 kg) in favour of the final 0.46 kg design because the lighter version cut the stress margin too far — a deliberate mass-vs-safety trade-off. Final design: 1.84 kg total mass (2.0 kg limit), 128 mm ground clearance, CG held within 8.6 mm of centre.

RELEVANT EXPERIENCE

Intern Trainee

Feb 2022 – Apr 2022

SASTRA University, India

- Worked across composites manufacturing, 3D printing, and aerodynamic wind-tunnel testing, correlating simulation predictions with physical test data.

Intern Trainee

Jun 2022 – Sep 2022

CIPET, Chennai, India

- Analysed industrial manufacturing and quality-control processes, building practical understanding of production-floor engineering standards.

Design Intern

Oct 2020 – Jan 2021

Space Technology and Aeronautical Rocketry (STAR), India

- Designed and simulated a model rocket and static test bed using OpenRocket and Proteus, gaining early hands-on exposure to aerospace R&D workflows.

Logistics Associate

Sep 2026 – Present

GXO Logistics, UK — promoted twice within a year (Warehouse Operator → Warehouse Associate → Logistics Associate); full-time role held in parallel with an active job search in aerospace engineering, not a career direction.

CERTIFICATIONS

- Ansys Certified Associate — Foundations in Fluid Dynamics
- Ansys Certified Associate — Foundations in Heat Transfer
- COMSOL Multiphysics Training — Aerothermal & Conjugate Heat Transfer
- COMSOL Multiphysics Training — Fluid-Structure Interaction for Aerospace Structures
- MATLAB Onramp & Simulink Onramp — MathWorks
- STAR-CCM+ for a Beginner — Flowthermolab
- Dassault Systèmes Certified SOLIDWORKS Professional (CSWP) — Mechanical Design
- Microsoft Certified: Azure Fundamentals (AZ-900) — Microsoft

ADDITIONAL INFORMATION

Right to work: UK Graduate visa — eligible to work full-time in the UK; sponsorship may be required for continued employment beyond the visa term. Notice period: 1 week. Languages: English (fluent), Tamil (native). Location: Leicester, UK — open to relocation across the UK and Europe.